ITA06 MACHINE LEARNING | ASSIGNMENT 1 (CO1)

Intelligent Early-Warning Decision Support System for Academic Risk Identification using Concept Learning and Decision Tree Learning

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Course: **ITA0604 – MACHINE LEARNING**

Notation. Attribute order for Dataset A is ⟨Attendance, Assignments, StudyHours, PriorCGPA, LMS, Mentoring⟩; Irr = Irregular, Pas = Passive, Reg = Regular. ⟨?×k⟩ means k consecutive "?". Entropies use $\log_2$ with 4 decimals. All numbers below were produced by my own Python code (ml.py, id3.py, tests.py) and agree with the hand traces.

Task 1 – Formulate the ML Problem

(a) T-P-E. Task T: classify a student, from week-6 attributes, as AtRisk = Yes/No. **Performance P:** recall on AtRisk = Yes (primary), with precision and F1 as secondary measures, on held-out students. **Experience E:** the labelled history of past students (Dataset A: 5, Dataset B: 14 examples) with end-of-semester labels from the academic council.

(b) Attributes.

Attribute	Type / values	Role	Why relevant
Attendance	Ordinal: Low, Medium, High	Input	Missed classes mean missed content; strongest behavioural early signal
Assignments	Binary: Irregular, Regular	Input	Shows discipline and keeps pace with the syllabus
StudyHours	Binary: Low, High	Input	Effort outside class (**risky**: self-reported, so biased and unverifiable)
PriorCGPA	Binary: Weak, Strong	Input	Past achievement predicts foundational readiness
LMS	Binary: Passive, Active	Input	Digital engagement; shows if materials are being used
Mentoring	Binary: No, Yes	Input	Support received (**also risky**: at-risk students are more likely to be enrolled, so it can be a confound)
AtRisk	Binary: Yes, No	Target	Council decision at end of semester

(c) Target concept. $c : X \rightarrow \{0,1\}$, with $c(x) = 1$ iff AtRisk = Yes. Instance space: $|X_A| = 3 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 =$ **96**, and 5 examples cover 5/96 = **5.2 %**; $|X_B| = 3 \cdot 2 \cdot 2 \cdot 2 =$ **24**, and 14 distinct examples cover 14/24 = **58.3 %**.

(d) Paradigm. Supervised learning (classification): every record carries a council-assigned label, the output is a discrete class, and performance can be measured against known labels.

(e) ML issues.

Issue	Evidence in these datasets	Mitigation
Small data / poor coverage	Dataset A covers 5.2 % of its space; B covers 58.3 %; all 6 test students are unseen combinations	Cross-validation; collect more records; prefer simple models
Label noise	Several advisors label; audit finds 5–10 % inconsistent (E6 contradicts E1)	Pruning, advisor audit, noise-tolerant stopping rule

Issue	Evidence in these datasets	Mitigation
Sampling bias	B contains only 14 of 24 combinations; Mentoring/StudyHours are self-selected	Stratified sampling; drop or down-weight risky attributes
Generalisation	Full tree: 100 % train but 50 % test (Task 8)	Depth/gain limits, held-out evaluation
Fairness and privacy	PriorCGPA may penalise weaker backgrounds; student records are personal data	Subgroup recall audit; access control; human review of flags

(f) Cost of errors. A false negative (missed at-risk student) loses the only chance for counselling, so the student may fail. A false positive costs only some counselling time. Hence **recall on AtRisk = Yes** is emphasised, while precision is monitored so advisors are not flooded.

Task 2 – Analyse the Learning Paradigm

Criterion	Supervised	Unsupervised	Reinforcement
Labels available	Yes: council's AtRisk	None used	Not labels; rewards only
Output	Class (Yes/No) per student	Clusters / profiles	Policy: action per state
Feedback signal	Error against true label	None (structure only)	Delayed reward (pass/fail, cost)
Data needed	Hundreds to thousands of labelled records	Unlabelled records, any amount	Many episodes of interventions
Evaluation	Recall, precision, F1 on held-out data	Cluster quality (subjective)	Cumulative reward
Interpretability	High with trees/rules	Medium: profiles need naming	Low to medium
Main risk	Learns advisors' label noise and bias	Clusters may not match risk	Cannot safely experiment on students

- **Unsupervised:** cluster students into behavioural profiles (for example "low attendance, passive LMS") to help advisors design programmes. It **cannot** say which profile will fail, because it never sees AtRisk, so it cannot give the flag decision.
- **Reinforcement (follow-up problem):** state = (attendance, assignments, LMS, CGPA band, interventions so far); action = {counselling, remedial class, mentoring, none}; reward = +1 if the student passes minus the intervention cost. It is unsuitable for initial classification because there is no sequential decision and no reward until semester end, and we cannot try "random" actions on real students.
- **Selected:** supervised learning. **Strongest counter-argument:** labels are subjective and noisy, so the model copies advisor bias. **Rebuttal:** only 5–10 % of labels are inconsistent, pruning tolerates this, and no unsupervised method could even be evaluated for the flagging task.
- **Hybrid:** cluster students first (unsupervised) and use clusters to audit label inconsistency or as an extra feature, then classify with a supervised tree. It improves label quality without losing interpretability.

Task 3 – Concept Learning Task (Dataset A)

Target concept c : X → {0,1}. Positive: **E1, E2, E4** (AtRisk = Yes); negative: **E3, E5**. Hypotheses are conjunctions of constraints (value, "?" or "∅"). Most general: **⟨?,?,?,?,?,?⟩**; most specific: **⟨∅,∅,∅,∅,∅,∅⟩**. E1 = ⟨Low,Irr,Low,Weak,Pas,No⟩ and E3 = ⟨High,Reg,High,Strong,Active,Yes⟩.

Hypothesis space. (i) Syntactically distinct: each attribute allows its values plus "?" and "∅", so $(3+2)(2+2)^5 = 5 \cdot 1024 =$ **5,120**. (ii) Semantically distinct: every hypothesis containing ∅ is the same empty concept, so $1 + (3+1)(2+1)^5 = 1 + 4 \cdot 243 =$ **973**. (iii) Instance space = **96**.

Ordering. h1 = ⟨Low,?,?,?,?,?⟩, h2 = ⟨Low,Irr,?,?,?,?⟩, h3 = ⟨?,?,?,Weak,?,?⟩. h1 ≥ h2, because every instance covered by h2 has Attendance = Low and so is covered by h1, but not conversely. h3 is **incomparable** with both: ⟨Low,Reg,·,Strong,…⟩ is covered by h1 but not h3, while ⟨High,Reg,·,Weak,…⟩ is covered by h3 but not h1 (and similar for h2).

Hypothesis	Covers	Consistent?	Reason
h_a = ⟨Low,?,?,?,Pas,?⟩	E1, E2, E4	**Yes**	Covers all positives, covers neither negative
h_b = ⟨?,Irr,?,?,?,?⟩	E1, E2, E4, **E5**	**No**	Covers negative E5 (Medium, Irregular, …)
h_c = ⟨Low,Irr,Low,?,?,?⟩	E1, E2	**No**	Fails to cover positive **E4** (StudyHours = High)

Satisfies vs consistent. Satisfying is about one instance: h_b satisfies E5 (it covers it). Consistent is about the whole training set: h(x) must equal c(x) for **every** example. h_b satisfies E1, E2, E4 correctly but also satisfies the negative E5, so it is not consistent.

Task 4 – Candidate Elimination on Dataset A (E1 → E5)

S_0 = ⟨∅×6⟩, G_0 = ⟨?×6⟩.

Step	Ex.	Label	Action taken	S after	G after	Reason
1	E1	Yes	Minimally generalise S	⟨Low,Irr,Low,Weak,Pas,No⟩	⟨?×6⟩	S_0 covers nothing, so S becomes E1; G covers E1 and is kept
2	E2	Yes	Generalise S	⟨Low,Irr,Low,?,Pas,No⟩	⟨?×6⟩	E2 differs only on PriorCGPA (Weak vs Strong), so it becomes "?"

Step	Ex.	Label	Action taken	S after	G after	Reason
3	E3	No	Specialise G	⟨Low,Irr,Low,?,Pas,No⟩ (unchanged)	⟨Low,?⁵⟩, ⟨?,Irr,?⁴⟩, ⟨?,?,Low,?,?,?⟩, ⟨?⁴,Pas,?⟩, ⟨?⁵,No⟩	S does not cover E3. Minimal specialisations of ⟨?×6⟩ use any value ≠ E3's: Attendance=Low/Medium, Assignments=Irr, StudyHours=Low, CGPA=Weak, LMS=Pas, Mentoring=No. **Rejected** (not more general than S): Attendance=Medium and CGPA=Weak. None of the 5 kept is less general than another
4	E4	Yes	Remove from G; generalise S	⟨Low,Irr,?,?,Pas,?⟩	⟨Low,?⁵⟩, ⟨?,Irr,?⁴⟩, ⟨?⁴,Pas,?⟩	E4 has StudyHours = High and Mentoring = Yes, so **removed** from G: ⟨?,?,Low,?,?,?⟩ and ⟨?⁵,No⟩. S generalises those two attributes to "?"
5	E5	No	Specialise the member of G covering E5	⟨Low,Irr,?,?,Pas,?⟩ (unchanged)	⟨Low,?⁵⟩, ⟨?⁴,Pas,?⟩	⟨Low,…⟩ and ⟨…Pas…⟩ do not cover E5, so they stay. ⟨?,Irr,?⁴⟩ covers E5 → candidates Attendance=Low/High, StudyHours=Low, CGPA=Weak, LMS=Pas, Mentoring=Yes. **Rejected** (not ≥ S): Attendance=High, StudyHours=Low, CGPA=Weak, Mentoring=Yes. Kept ⟨Low,Irr,?⁴⟩ and ⟨?,Irr,?,?,Pas,?⟩ are then **removed** as less general than ⟨Low,?⁵⟩ and ⟨?⁴,Pas,?⟩

Final boundaries: S_5 = {⟨Low, Irr, ?, ?, Pas, ?⟩}; G_5 = {⟨Low,?,?,?,?,?⟩, ⟨?,?,?,?,Pas,?⟩}.

Task 5 – Version Space and Inductive Bias

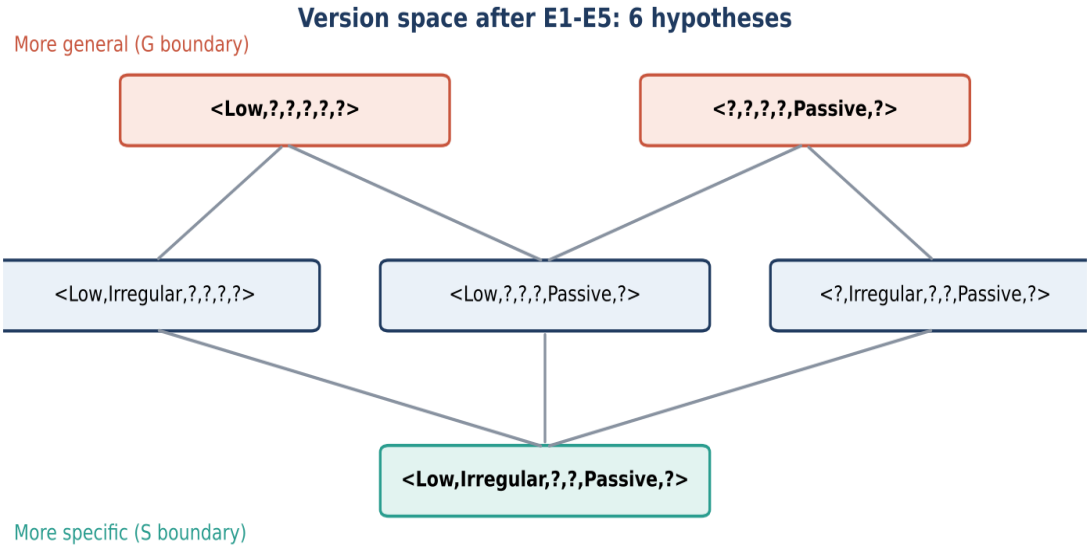


Figure 1: Version space lattice between S_5 (bottom) and G_5 (top)

Version space (6 hypotheses): h1 = S_5 = ⟨Low,Irr,?,?,Pas,?⟩; h2 = ⟨Low,Irr,?,?,?,?⟩; h3 = ⟨Low,?,?,?,Pas,?⟩; h4 = ⟨Low,?,?,?,?,?⟩ (G); h5 = ⟨?,Irr,?,?,Pas,?⟩; h6 = ⟨?,?,?,?,Pas,?⟩ (G). **⟨?,Irr,?,?,?,?⟩ is not in it** because it covers the negative E5 and is not more specific than any member of G_5.

Query	h1	h2	h3	h4	h5	h6	Vote	Verdict
Q1 = ⟨Low,Reg,High,Weak,Pas,No⟩	No	No	Yes	Yes	No	Yes	3 Yes : 3 No	**Ambiguous**
Q2 = ⟨Low,Irr,Low,Strong,Pas,Yes⟩	Yes	Yes	Yes	Yes	Yes	Yes	6 Yes : 0 No	Unanimous: At risk

Q1 is ambiguous because Assignments = Regular rejects the hypotheses that require Irregular, while Low/Passive satisfy the others. The **most informative next query** is one that splits the version space 3:3, so any answer removes half of it. Eight instances do this, all of the form ⟨Low,Reg,·,·,Pas,·⟩; for example ⟨Low,Reg,Low,Strong,Pas,No⟩ (Q1 itself also qualifies).

Concept	Meaning in this problem
Training data	E1–E5 with labels
Hypothesis space	All 973 semantically distinct conjunctions over the six attributes
Version space	The 6 hypotheses in H consistent with E1–E5
Inductive bias	Assumption that the target is a single conjunction of attribute constraints

Bias and its effect. The bias of Candidate Elimination is "the target concept is a conjunction in H". Adding the positive ⟨High,Reg,High,Strong,Pas,No⟩: S becomes ⟨?,?,?,?,Pas,?⟩ and ⟨Low,?⁵⟩ is removed from G (it does not cover the new example), so the version space shrinks from **6 to 1** hypothesis ("LMS = Passive"); the Attendance = Low evidence is lost because a conjunction cannot say "Low OR passive". With the **power set** as hypothesis space there are $2^{96} \approx \mathbf{7.9 \times 10^{28}}$ concepts; S would be just the disjunction of the positives and G the negation of the negatives, so every unseen instance gets exactly half the votes: with no bias the learner cannot generalise.

Task 6 – Decision Tree by ID3 (Dataset B)

Entropy of the full set: 7 Yes, 7 No → $H(S) = -(½)\log_2½ - (½)\log_2½ =$ **1.0000**. **Root split (all four attributes):**

Attribute	Subsets (size; Yes/No; entropy)	Weighted entropy	Gain
Attendance	High 4; 1/3; 0.8113 | Low 5; 4/1; 0.7219 | Medium 5; 2/3; 0.9710	0.2318+0.2578+0.3468 = 0.8364	0.1636
Assignments	Irregular 8; 5/3; 0.9544 | Regular 6; 2/4; 0.9183	0.5454+0.3936 = 0.9389	0.0611
LMS	Passive 7; 5/2; 0.8631 | Active 7; 2/5; 0.8631	0.4316+0.4316 = 0.8631	0.1369
PriorCGPA	Strong 7; 1/6; 0.5917 | Weak 7; 6/1; 0.5917	0.2959+0.2959 = 0.5917	**0.4083**

Root = PriorCGPA (highest gain). Impure branches:

Node	Size; Yes/No; entropy	Gain: Attendance / Assignments / LMS	Choice
CGPA = Strong	7; 1/6; 0.5917	0.3060 / 0.1281 / 0.1981	Attendance
CGPA = Weak	7; 6/1; 0.5917	0.3060 / 0.1981 / 0.1981	Attendance
Strong ∧ Att = Low	2; 1/1; 1.0000	– / 1.0000 / 1.0000 (**tie**)	Assignments (tie rule)
Weak ∧ Att = High	2; 1/1; 1.0000	– / 1.0000 / 1.0000 (**tie**)	Assignments (tie rule)

Stopping criteria: node is pure, or no attributes remain, or gain = 0. **Tie-breaking:** the attribute that appears first in column order (Attendance, Assignments, LMS, PriorCGPA); a tied leaf is labelled Yes (false negatives are costlier).

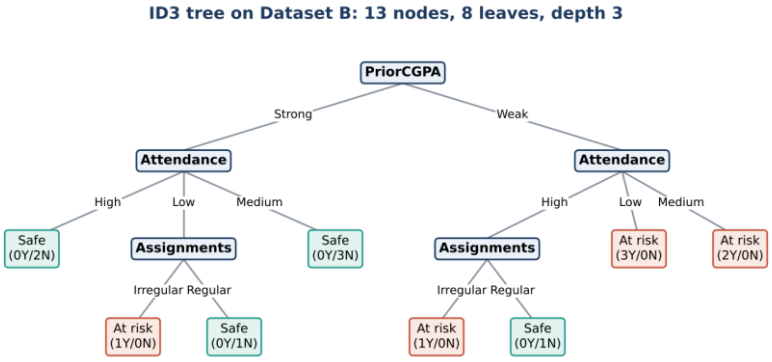


Figure 2: Final ID3 tree (M2)

Rules. R1: IF CGPA=Strong ∧ Att=High THEN Safe. R2: Strong ∧ Low ∧ Assignments=Irr → **At risk**. R3: Strong ∧ Low ∧ Reg → Safe. R4: Strong ∧ Medium → Safe. R5: Weak ∧ High ∧ Irr → **At risk**. R6: Weak ∧ High ∧ Reg → Safe. R7: Weak ∧ Low → **At risk**. R8: Weak ∧ Medium → **At risk**. **Statistics:** depth 3, 13 nodes, 8 leaves (5 decision nodes), training accuracy 14/14 = **100 %**.

Task 7 – Heuristic Search in Decision Tree Learning

Search formulation. Start state: a single empty leaf. Operator: replace a leaf by a test on one unused attribute (one branch per value). Heuristic: information gain (greedy, hill-climbing). Termination: pure node, no attributes left, or zero gain. In my tree the root evaluated 4 candidates, the two level-1 nodes 3 each, and the two level-2 nodes 2 each, giving **4 + 6 + 4 = 14 gain computations** in total. ID3 searches the

space of all Boolean functions on 24 points (2^{24} = 16,777,216), far larger than the 973 conjunctions of Task 3, so its **representational bias is weak** and the bias comes only from the **search preference** for short trees with high-gain attributes near the root; Candidate Elimination instead has a strong **restriction bias**.

Root choice	Info gain	Split info	Gain ratio
Attendance (3 values)	0.1636	1.5774	0.1037
PriorCGPA (2 values)	0.4083	1.0000	0.4083
StudentID (14 values)	**1.0000** (14 pure singletons, weighted entropy 0)	$\log_2 14 = 3.8074$	0.2627

Gain ratio divides by split information, which grows as $\log_2(k)$ with the number of branches, so many-valued attributes are penalised. StudentID gets the maximum gain of 1.0 but only by memorising each student; it cannot classify any new one. I adopt **gain ratio** (it agrees with gain here: PriorCGPA), as protection against such attributes.

Tree	Root	Depth	Nodes	Leaves	Train acc.	Test acc.
Greedy ID3	PriorCGPA	3	13	8	100 %	50.0 %
Alternative	Attendance	3	14	8	100 %	33.3 %

The greedy choice gave the smaller (13 vs 14 nodes) and better-generalising tree, though this is not guaranteed. ID3 never backtracks because re-evaluating earlier choices would need an exponential search. **Advantages:** (1) cheap: 14 gain computations instead of enumerating trees; (2) it found the dominant PriorCGPA structure at the root, giving readable rules. **Limitations:** (1) an early split cannot be undone, and a poor one propagates (the Attendance-rooted tree needs an extra node); (2) ties such as Assignments vs LMS (both 1.0000) are broken arbitrarily, and with 14 examples a single record can change a split.

Task 8 – Overfitting and Underfitting

Model	Nodes / leaves	Train acc.	Test acc.	Gap	Diagnosis
M0 (no split, majority)	1 / 1	7/14 = 50.0 %	3/6 = 50.0 %	0	**Underfits** (too simple, high bias)
M1 (depth 1)	3 / 2	12/14 = 85.7 %	5/6 = 83.3 %	2.4 pts	**Best generalisation**
M2 (full tree)	13 / 8	14/14 = 100 %	3/6 = 50.0 %	50 pts	**Overfits** (memorises)

Causes. Only 14 training students cover 58 % of the 24 combinations and all 6 test students are unseen combinations; several leaves hold one or two students, so each leaf is an anecdote, and any label noise is memorised by the deep tree.

Control. (1) Maximum depth = 1 → M1 (below): train 85.7 %, test **83.3 %**. (2) min_samples_split = 4 → depth 2, 9 nodes, 6 leaves, train 85.7 %, test 50.0 %; this alone is not enough because 7-sample nodes can still split. (3) Minimum gain $\theta = 0.35$: the root (0.4083) splits but the best level-1 gain (0.3060) is below θ, giving the same tree as M1. Tied leaves are labelled Yes. **Chosen setting: depth 1**; leave-one-out on the 20 labelled records gives 85 % for depth 1–2 against 55 % for the full tree. Six test examples are very few (one error = 16.7 points), so I would use repeated cross-validation and collect more data.

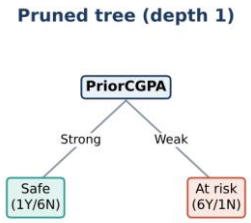


Figure 3: Pruned tree M1 (PriorCGPA stump)

Task 9 – Implementation and Testing

Own Python code (no library for CE or ID3): **ml.py** (hypotheses with "?"/"∅", covers(), more-general-than(), Candidate Elimination with S/G sets, prints S and G after every example, detects an empty version space); **id3.py** (entropy, gain, gain ratio, recursive ID3, max-depth / min-samples / min-gain controls, prediction, rule printing, metrics); **exp.py** (experiments, fixed seed 42); **tests.py** (TC1–TC8); **README.md** (run steps: python3 tests.py). The files are provided with this document.

Test	Input	Expected	Observed	Result
TC1	CE on Dataset A	S, G match Task 4 trace	S_5 = ⟨Low,Irr,?,?,Pas,?⟩; G_5 = {⟨Low,?⁵⟩, ⟨?⁴,Pas,?⟩}	Pass
TC2	Pure set; 7Y/7N set	0 and 1.0000	0.0000, 1.0000	Pass
TC3	Root gains, Dataset B	0.1636, 0.0611, 0.1369, 0.4083	Identical	Pass
TC4	Tree on B, 14 examples	14/14 correct, 8 rules	14/14 correct, 8 rules	Pass
TC5	Test Set T, full tree	Yes,Yes,No,Yes,No,Yes (50 %)	Same predictions, 50 %	Pass
TC6	E6 contradicts E1	Report empty version space, no crash	"VERSION SPACE EMPTY" printed	Pass
TC7	Attendance = "Very Low"	Policy: fall back to node majority class	Predicted Yes, no crash	Pass
TC8	Tie (Assignments vs LMS)	Deterministic: first in column order	Same tree on repeat runs	Pass

Task 10 – Experimental Analysis and Recommendation

10A. Experiments

Dataset C is synthetic (seed 42, 600 records, 4 attributes, configurable label-noise rate). Rule: AtRisk = Yes if (PriorCGPA=Weak ∧ (Attendance≠High ∨ Assignments=Irr)) ∨ (PriorCGPA=Strong ∧ Attendance=Low ∧ Assignments=Irr ∧ LMS=Pas). 400 records train, 200 validate.

Dataset	Examples	Attributes	Yes / No	Cardinalities	Space coverage
A	5	6	3 / 2	3,2,2,2,2,2	5/96 = 5.2 %
B	14	4	7 / 7	3,2,2,2	14/24 = 58.3 %
T	6	4	3 / 3	3,2,2,2	6 new combos (B+T: 20/24 = 83.3 %)
C	600	4	262 / 338	3,2,2,2	24/24 = 100 %

Evaluation (Yes = positive)	Accuracy	Precision	Recall	F1	TP / FN / FP / TN
Full tree on T (train B)	50.0 %	0.500	0.667	0.571	2 / 1 / 2 / 1
Depth-1 tree on T (train B)	83.3 %	0.750	1.000	0.857	3 / 0 / 1 / 2
Full tree, leave-one-out (20 records)	55.0 %	0.545	0.600	0.571	6 / 4 / 5 / 5
Depth-1 tree, leave-one-out (20 records)	85.0 %	0.818	0.900	0.857	9 / 1 / 2 / 8

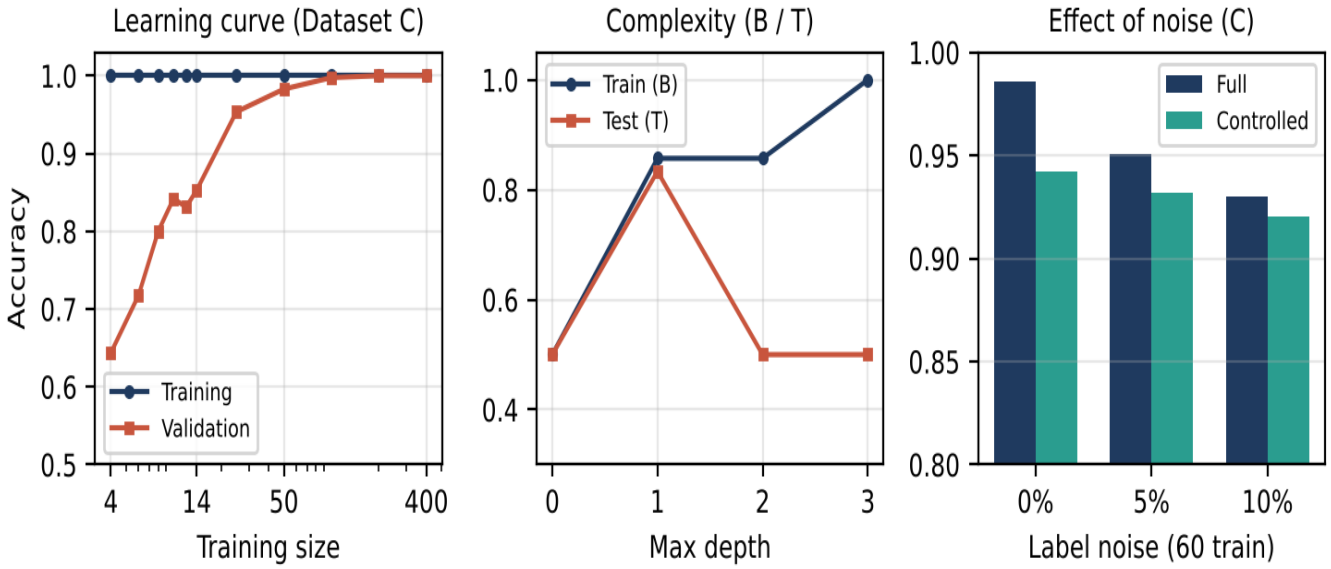


Figure 4: Learning curve, model complexity (B/T) and label-noise effect

Max depth	Nodes / leaves (B)	B train / T test	Nodes (C)	C train / validation
0	1 / 1	50.0 % / 50.0 %	1	57.0 % / 55.0 %
1	3 / 2	85.7 % / 83.3 %	3	86.3 % / 86.5 %
2	9 / 6	85.7 % / 50.0 %	9	90.5 % / 90.0 %
3 (full on B)	13 / 8	100 % / 50.0 %	13	97.0 % / 97.0 %
4 (full on C)	13 / 8	100 % / 50.0 %	15	100 % / 100 %

Learning curve: on B the validation accuracy stays near 50 % for 4–14 examples because every training size is memorised (train 100 %); on C validation rises from 64 % (4 examples) to 85 % (14), 95 % (25) and 100 % (200+). **Noise (60 training records, 20 repeats):** 0 / 5 / 10 % noise → full tree accuracy 98.6 / 95.1 / 93.0 % with 13.0 / 21.0 / 24.9 nodes; controlled tree (min split 10, min gain 0.02) 94.2 / 93.2 / 92.0 % with 10.2 / 13.4 / 14.8 nodes. With 400 records both stay ≥ 99.7 %, but at 10 % noise the full tree grows to 43.4 nodes against 22.6 for the controlled tree. **CE vs tree on B:** CE on B's four attributes processes examples 1–3 (S = ⟨Low,?,?,?⟩) and the **version space collapses at example 4** (Low, Regular, Active, Strong = No lies inside S).

Aspect	Candidate Elimination	Decision tree
Representable	One conjunction only (no OR)	Any Boolean function
Noise	One inconsistent record empties the version space	Tolerated by majority leaves and pruning
Unseen values	Cannot match; may abstain	Falls back to node majority

Aspect	Candidate Elimination	Decision tree
Data needed	Few, but consistent	More, but robust
Result on B	Collapsed at example 4	100 % train; 83–85 % when pruned

10B. Theory vs Experiment

- **Overfitting and collapse confirmed:** the full tree gave 100 % train / 50 % test and the version space collapsed on B (example 4), as predicted in Tasks 5 and 8.
- **Dataset characteristics:** small size and 58 % coverage drive the overfitting; more data (C) removed the gap, and noise inflated tree size more than it reduced accuracy.
- **Inductive bias:** B's concept is disjunctive (Strong ∧ Low ∧ Irr is at risk while Strong ∧ Low ∧ Reg is not), so the conjunctive bias fails and the unrestricted tree bias succeeds. **Heuristic:** gain and gain ratio agree on PriorCGPA; they would differ only for many-valued attributes such as StudentID. **Complexity:** depth 1–2 is best on small data.
- **Limitations and threats to validity:** 6 test examples, C is synthetic with a clean deterministic rule, and one random split.

Recommendation

Use a **pruned/controlled decision tree** (depth ≤ 2, gain ratio, tied leaves → Yes). Evidence: depth-1 tree gives recall 0.90, precision 0.82 and accuracy 85 % in leave-one-out, against a CE collapse on B and 55 % for the unpruned tree; trees also give readable IF–THEN explanations, tolerate noise and handle unseen cases through node majority. **Candidate Elimination would be preferable** only if the data were small, noise-free and truly conjunctive, or when we want to choose the next student to label (query selection), as in Task 5.

Task 11 – Day-4 Design Challenge

A. Impact Analysis

Change	Assumption violated	Affects	Consequence
≈2,000 labelled records, weekly updates	Small data	Data / evaluation	Trees can be properly validated; CE's rare consistency becomes more fragile
5–10 % inconsistent labels	Noise-free labels	Data / algorithm	CE collapses; unpruned tree memorises noise
New values (Attendance = Very Low)	Fixed attribute values	Hypothesis space	CE has no way to match; tree needs a policy
Recall ≥ 90 %, explanations required	Accuracy-only evaluation	Evaluation	Need threshold/weighting and rule paths
Disjunctive risk patterns	Conjunctive concept	Hypothesis space	CE cannot represent the target; trees can

E6 vs N2. E6 has exactly the same attributes as E1 but the opposite label, so a method that assumes consistent data has **no consistent hypothesis** and fails outright. N2 (High, Regular, Active, Strong → At risk) contradicts no record but is an outlier against the pattern "strong, high attendance, active ⇒ safe": an unpruned tree builds extra splits just for it, whereas a noise-tolerant method treats it as noise and absorbs it into a majority leaf.

B. Re-evaluation of the Original Design (run on my code)

Experiment	Observed result
CE: E1–E5, then **E6**	S and G both become empty at E6: **version space collapses**
CE: E1–E5, then **E7**	$S_7 = G_7 = \langle ?,?,?,?,Pas,? \rangle$; version space shrinks from 6 to 1 hypothesis
CE: then **E8** (cumulative)	E8 is positive but G does not cover it (LMS = Active): **collapse at E8**
Tree + N1–N3	13 → **18 nodes**, 8 → **11 leaves**, depth 3 → **4**; training accuracy still 100 %. New branch Attendance = Very Low (N1); N2 forces extra splits (Strong → High → Assignments → LMS)
Original tree on N1–N3	N1: Yes (unseen value → node majority 6Y/1N, correct); N2: No (**missed**); N3: Yes. Recall 2/3
Recall ≥ 90 % (LOOCV, 23 records)	Full tree recall 46.2 %, accuracy 43.5 %; depth-1 tree recall **84.6 %**, precision 84.6 %, accuracy 82.6 %. **Not met** on this small sample
Explanations	Tree: root-to-leaf rule. CE: none once the version space is empty

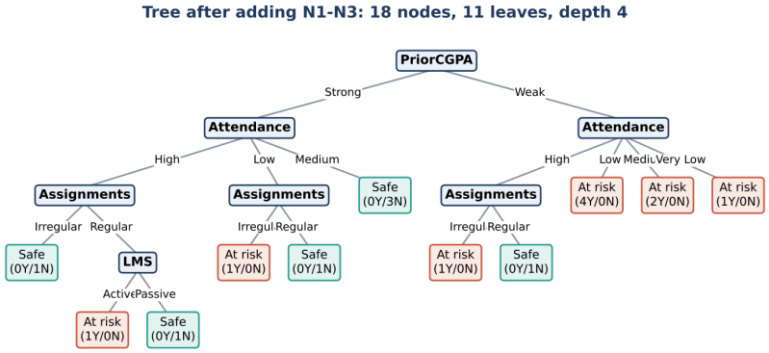


Figure 5: Tree rebuilt after adding N1–N3

C. Decision: MODIFY

Keep decision-tree learning as the core; demote Candidate Elimination to an offline **data-audit** tool (an empty version space flags contradictory records for advisors). Changes:

- **Tree induction:** gain ratio; stopping rule = node size < 20 or gain < 0.02; reduced-error pruning on a 20 % validation split; retrain weekly.
- **Unseen values/combinations:** any new value is routed to an "Unknown" branch that uses the parent node's class distribution and adds the flag "needs advisor review"; new combinations are handled by the tree's generalisation.
- **Recall requirement:** predict At risk if the leaf's P(Yes) ≥ τ = 0.35 (equivalent to weighting Yes errors about twice). On Dataset C (10 % noise, controlled tree) this moved recall 0.923 → 0.942 and precision 0.916 → 0.885: more students are caught at the price of more false alarms.
- **Explanations:** show the rule path and leaf counts, e.g. "PriorCGPA = Weak ∧ Attendance = Low: 4 of 4 similar past students were at risk".

Pseudocode: train(D): clean D (audit with CE); T = ID3(D, gain ratio, min_size 20, min_gain 0.02); prune T on validation set; for each student x: p = leaf_distribution(T, x) (unknown value → parent distribution); flag if p(Yes) ≥ 0.35; explain with the rule path.

D. Evidence-Based Defence

Bias: the target is disjunctive, so the conjunctive bias of CE is wrong, and CE collapsed three times (E6, E8, and B). **Search:** greedy gain-ratio search finds the PriorCGPA structure cheaply and resists many-valued attributes. **Fit:** the full tree (50 % test) overfits, while a pruned tree reached 83–85 %.

Experiments: pruned trees stay accurate under 10 % noise (92 % at 60 records, ≥ 99 % at 400) with half the nodes. **Recall:** on the 23 real records the best measured recall was 84.6 % and **the 90 % target is not yet demonstrated**; on the larger Dataset C the controlled tree reached 92–94 % at 60 records and 98–100 % at 400, so the target is realistic with ~2,000 records.

Remaining limitation: Dataset C is synthetic and the 23 real records are few; also behaviour may drift as new study modes appear. **Monitoring:** every week compare flags with end-of-semester labels and track recall (alert if < 90 %), precision, the share of "Unknown" values (alert if > 5 %), subgroup recall, and an advisor audit of a sample of labels; retrain when an alert fires.